

ANT-EV Setup and Operation Guide

Simple prerequisites, running steps, model status, and safety boundary

This project has one purpose: demonstrate how PRISM-ANT safely splits a station's limited power among EVs. The interface shows the assigned kW and a plain reason for every active vehicle.

Install once

Install	Why
Node.js LTS with npm	Runs the React/Vite web application.
Python 3.10+ (optional)	Runs training utilities and document-generation tools.
Git (optional)	Source control.

Run the software

1. Open a terminal in the project root.
2. Run **npm install**.
3. Run **npm run dev**.
4. Open the local URL printed in the terminal.
5. Select a demo, add a vehicle, then use Run demo or Step.

What to test

Load the 10-vehicle scenario. Add an EV with a short deadline. Mark it as emergency. Change its temperature. Complete a vehicle. Each action should immediately change the safe cap, readiness reserve, and remaining power split.

Build check

Run **npm run build**. A successful build checks the TypeScript front end. It does not certify a physical charging installation.

Model status and training

The interface currently uses local prototype estimators through one M1-M6 signal contract. M1/M2 datasets and scripts are under **training/**. Do not describe M3-M6 as trained production models until suitable datasets, validation, and artifact integration are complete.

Model	Training goal	PRISM-ANT handoff
M1	Energy and charging duration	Need, readiness risk, confidence
M2	Station demand forecast	Forecast pressure
M3	Battery health	Health risk and confidence
M4	Thermal/charging stress	Safe cap and thermal risk
M5	Future SOC/temperature	Future-temperature risk
M6	Strategy ranking	Safe strategy value

Safety boundary

Use a simulated environment until charger command acknowledgement, OEM battery limits, BMS data freshness, electrical protection, cybersecurity, emergency procedures, and professional safety review are in place. A trained model must not replace hard safety constraints.

Documentation set

This folder intentionally contains only three PDFs: this setup guide, the PRISM-ANT core algorithm, and the PRISM-ANT patent specification.

PRISM-ANT in the Running Demo

Use these checks while demonstrating the system.

Demo action	Expected PRISM-ANT response
Add an EV with a short deadline	Its readiness reserve and priority increase, subject to its safe cap.
Add an emergency EV	Its urgency increases; its safe cap still controls the maximum kW.
Raise battery temperature	M4/M5 lower the safe cap before power sharing.
Complete an EV	Its reserve becomes zero and its released kW is shared by remaining EVs.
Change station demand or deadline	The forecast-deadline interlock and reserve are recalculated.

What to explain to an examiner or reviewer

For each EV, point to: its energy need, deadline, safe cap, readiness reserve, final kW, and the short reason. This demonstrates that PRISM-ANT is a traceable, safety-first allocation method rather than a black-box score.

Complete ANT-EV Software Explanation Guide

One-sentence description: ANT-EV is a station-level EV charging decision system. It uses model insights and PRISM-ANT to split limited charging power safely, explain the reason for every allocation, and reallocate immediately when station conditions change.

1. What the software does

Part	Purpose
Charge Split screen	Shows every active EV, its allocated kW, safe cap, readiness reserve, and live reason.
Add Vehicle	Lets the user add SOC, target, deadline, maximum intake, temperature, and medical/goods urgency to test dynamic reallocation.
Models screen	Shows each M1-M6 model input, output, exact parameter sent to PRISM-ANT, and decision use.
Emergency screen	Shows why medical or goods urgency changes service order while safety limits remain active.
Architecture screen	Shows the live data path: request -> models -> safety mask -> PRISM-ANT -> power command -> explanation.
Patent screen	Explains the candidate PRISM-ANT technical claim focus, not a legal guarantee.

2. Simple data flow

Vehicle request: SOC, target SOC, deadline, maximum intake, temperature, urgency, and station capacity.

Models: M1-M6 convert these inputs into energy need, forecast, health, thermal, future-state, and strategy signals.

PRISM-ANT: applies the safety cap, calculates readiness reserve, shares remaining power, and saves the explanation.

Output: a safe kW command and a plain reason for each EV.

3. What each model contributes

Model	Plain purpose	Parameter to PRISM-ANT
M1 - Energy Need	Estimates how much energy and time an EV needs.	energy need, readiness risk, confidence
M2 - Demand Forecast	Estimates upcoming station pressure.	forecast pressure
M3 - Battery Condition	Estimates battery-health caution.	health risk, confidence
M4 - Charging Stress	Finds safe power from temperature and C-rate.	thermal risk, safe power cap
M5 - Future State	Checks whether a power decision becomes unsafe later.	future temperature risk
M6 - Strategy Value	Ranks only the already-safe choices.	strategy value

4. How PRISM-ANT makes a decision

Step	Explanation
Safety mask	Find the maximum safe kW. No EV can exceed it, including emergencies.
Readiness reserve	Calculate the minimum safe kW needed to reach the requested target before departure.
Fair remaining share	Split all remaining safe station kW by priority without exceeding any cap.
Live recalculation	Repeat after a vehicle arrives, completes, heats up, changes deadline, or becomes emergency.
Explanation	Show safe cap, reserve, final kW, models used, and short reason.

5. How ANT-EV differs from common approaches

Common approach	Problem	ANT-EV / PRISM-ANT difference
First come, first served	Arrival order can ignore deadlines.	Calculates a readiness reserve from energy need and time left.
Equal power split	All vehicles receive the same kW despite different risks and safe limits.	Uses individual safe caps, needs, deadlines and remaining-power share.
Single priority queue	Safety can be hidden inside a score.	Safety mask is applied first and cannot be overridden.
Opaque AI allocation	User cannot see why an EV received a value.	Shows model signals, reserve, cap, priority reason and assigned kW together.

6. Demonstration script

Start: “This station has limited capacity. PRISM-ANT decides how to share it safely.”

Show a normal EV: “Its target and deadline create an energy need and readiness reserve.”

Add an urgent EV: “The station recalculates. The urgent EV receives more priority, but only inside its safe cap.”

Raise temperature: “M4 and M5 lower the cap. Safety happens before priority.”

Complete an EV: “Its reserve becomes zero. PRISM-ANT immediately gives released power to other safe EVs.”

Open Models: “Each model has a named output, not an unexplained black-box score.”

Open Patent Spec: “The candidate invention is the ordered method: model evidence -> safety mask -> readiness reserve -> fair sharing -> event record.”

7. Patent discussion - careful wording

Say: “PRISM-ANT is our proposed technical claim focus.” Do not say: “It is already patented” or “no one has done this.” Dynamic allocation, SOC, temperature, deadlines and priority are known ideas. The possible differentiation is the exact ordered workflow, named model evidence, forecast-deadline interlock, narrow emergency rule, and explanation linked to the actual command.

8. Important operating boundary

This is a demo and decision-support design. Physical chargers require OEM BMS limits, charger acknowledgements, electrical protection, telemetry freshness checks, cybersecurity controls, fail-safe behaviour, and professional safety review. The algorithm must not be used as the only safety control.